Chapter 6

Sellers and Incentives

ECON 003
UC Riverside · Summer 2026



Where We Left Off

In Chapter 5 you built the demand curve out of one buyer's problem. Today you build the supply curve out of one seller's problem.

THE BUYER'S PROBLEM

A budget line, a bang-per-buck rule, and a quantity for every price. Stack those up and you have a demand curve.

THE SELLER'S PROBLEM

A production function, a cost curve, and a quantity for every price. Stack those up and you get a supply curve.

SAME MACHINERY

Optimise at the margin. Keep going while the next unit adds more than it costs. Stop when they are equal.

Chapter 4 asked you to trust the supply curve. Today you build it.

Today's Roadmap

1. Perfect competition

2. The seller's problem

3. To the supply curve

4. Producer surplus

5. Short run to long run

PART 1

Sellers in a Perfectly Competitive Market

Competitive Market Review

A perfectly competitive market: many small buyers and sellers, an identical product, and free entry and exit.

NOBODY IS BIG ENOUGH

So many buyers and sellers that no single one of them can move the market price by anything they do.

THE PRODUCT IS IDENTICAL

One seller's unit is a perfect substitute for another's.



ANYONE CAN COME AND GO

No license, no patent, no barrier.



This is a benchmark, not a description. Very few real markets satisfy all three.

Price Takers — Quantity Is Your Only Lever

A price taker treats the market price as given. It is a fact about the world, not a decision the firm gets to make.

WHAT THE FIRM DECIDES

Only ONE thing: how much to produce. Price is handed to it by the market and it takes that number as given.

WHAT HAPPENS IF IT TRIES

Charge a bit more and you sell nothing — every buyer goes to an identical seller. Charge a bit less and you have thrown money away for no reason, because you could have sold everything at the market price.

Identical Goods — and Why That Matters

If nobody can tell one seller's unit from another's, nobody can charge more than anybody else.

Free Entry and Free Exit



Nothing legal or technical stops a new firm setting up — or stops one leaving.

ENTRY FOLLOWS PROFIT

Profit is a signal, and in this model everybody can see it and anybody can act on it.

EXIT FOLLOWS LOSS

Firms losing money leave. Both movements push the market back toward the same place — which is what Part 6 is about.

Image: Acemoglu, Laibson & List, Microeconomics 3e

PART 2

The Seller's Problem

What Nick Built

Three decisions — and they are the three this whole section is about.

1. PRODUCTION

One jumbo-pop in, dozens of small ones out. Melt, re-freeze, add a stick. That is a production function.





2. COSTS

One Jumbo-pop turns into hundreds of Pawpsicles. Plus the stick and the melted syrup.



3. REVENUE

A queue of hamsters outside a bank.



Seller’s Problem

The seller's goal is to maximize profit. Getting there means answering three questions, in order.



…and the three together answer one question: how much should I make?

Starting a McDonald’s Franchise



CAPITAL INVESTMENT

A $45,000 franchise fee, and $1.47 to $2.73 million for equipment, seating, signage and opening stock. At least $750,000 of it has to be your own money, not borrowed.



TIME

Six to twelve months of training before you are even considered for a restaurant, and an agreement that usually runs twenty years.

McDonald's 2025 Franchise Disclosure Document, Items 5 and 7; mcdonalds.com/us/en-us/us-franchising (costs, process and FAQ pages), retrieved 9 August 2026.

Question 1: How Do You Make It?

Production is the process that turns inputs into outputs. Physical capital is any good — machine, building, fryer — used to do it.

INPUTS



Labor, capital, land, raw materials.

THE TECHNOLOGY



The recipe: how much output a given bundle of inputs yields.

OUTPUT



What comes out and gets sold. Meals served.

How do you make a burger? Beef, a bun, a grill, and somebody who knows how.

Short Run and Long Run

The short run is any period in which at least one input cannot be changed. The long run is the period in which all of them can.

SHORT RUN — THE KITCHEN IS FIXED

Serving lunch today, you are stuck with the grills and the floor space you already have.



LONG RUN — NOTHING IS FIXED

Given enough time you can add a second drive-through lane, take the unit next door, or close the restaurant entirely.



Fixed and Variable Factors

FIXED FACTOR OF PRODUCTION

An input that cannot be changed in the short run, and that stays the same no matter how much you produce. The building. The fryers.





VARIABLE FACTOR OF PRODUCTION

An input you can change within the period, and that moves with the level of output. Crew hours, beef, buns, power.

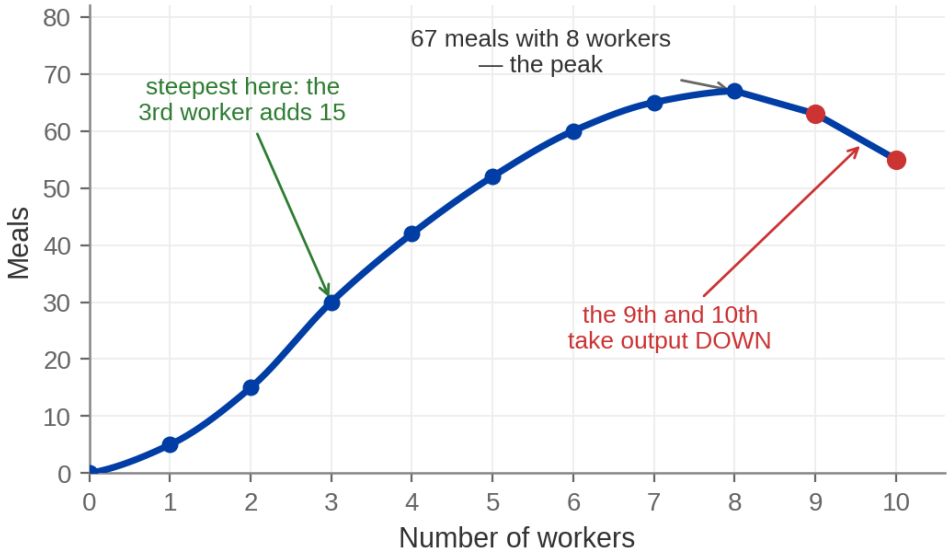






Production at McDonald's

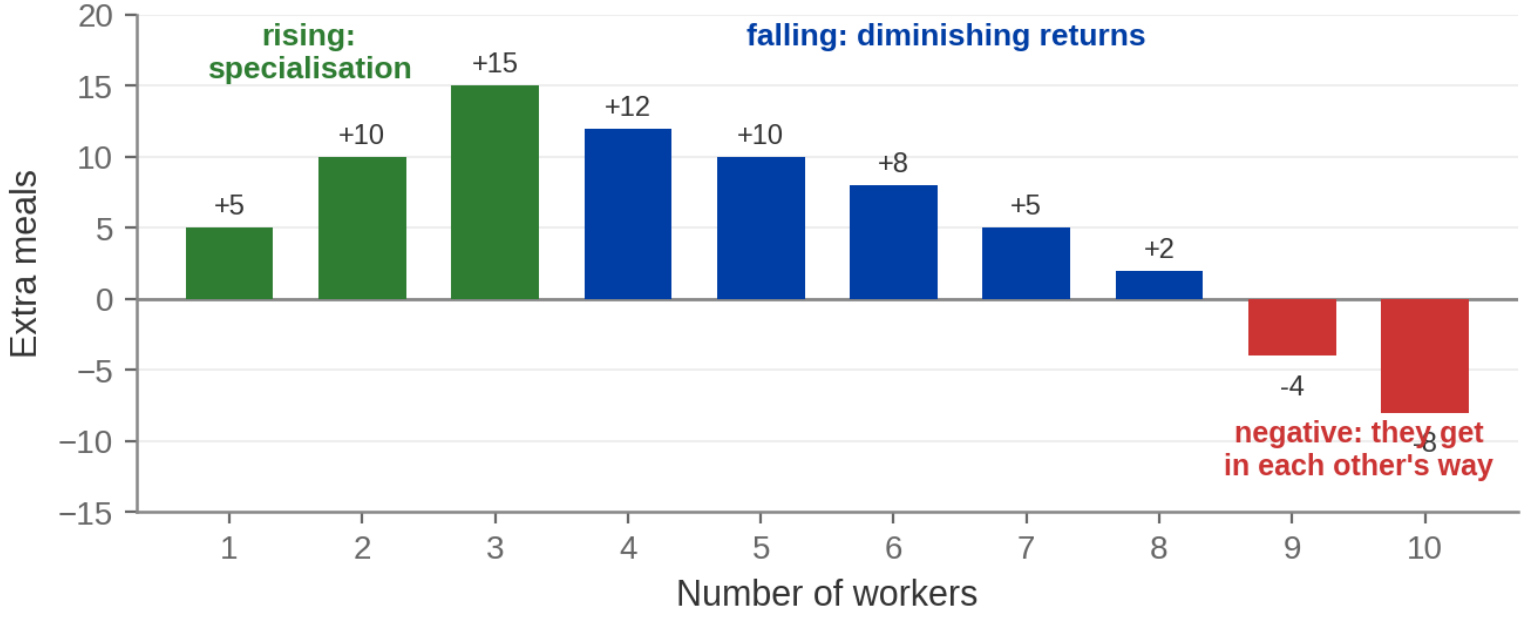
Workers	Meals	Marginal product
0	0	—
1	5	+5
2	15	+10
3	30	+15
4	42	+12
5	52	+10
6	60	+8
7	65	+5
8	67	+2
9	63	−4
10	55	−8



One kitchen, one set of grills, and workers added one at a time.

Marginal Product: Rise, Fall, Negative

Marginal product = the change in total output from adding one more worker.



Three Things This Table Says

1. IT RISES AT FIRST — SPECIALISATION

One person doing everything wastes time switching tasks. Two can split the work and each get good at half of it. That is why workers 2 and 3 each add more than the one before.

2. THEN IT FALLS — DIMINISHING RETURNS

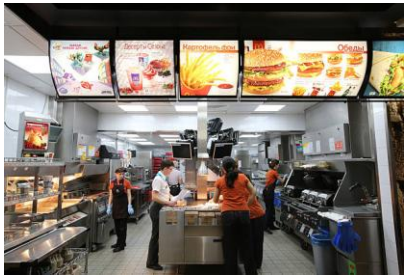
At some point each extra worker adds less than the one before. Not because he is worse, but because the kitchen he works in is fixed and there is less of it to go round.

3. IT CAN GO NEGATIVE

Keep adding people to the same kitchen and eventually they get in each other's way. The 9th worker takes output DOWN by 4 meals.

The law of diminishing returns: with at least one input fixed, each additional unit of a variable input eventually adds less than the one before it.







Question 2: What Does It Cost?

Total cost = variable cost + fixed cost.
Which is which depends on the length of the run.

FIXED COST (FC)



The cost of the fixed factors. The same at zero meals as at a hundred — the rent, the fryers, the insurance, the loan. Also called overhead. Here: $100 an hour.

VARIABLE COST (VC)

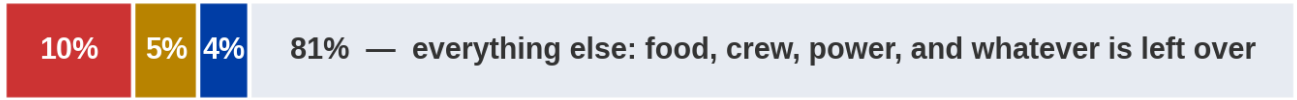


The cost of the variable factors. Moves with output — the crew on shift, the beef, the buns, the power for the grills.

What It Actually Costs to Run One

$3,966,000 of gross sales in an average year

Rent 10% (illustrative) | Ad fund 4%



Royalty 5%

19% is gone before a single burger is paid for



Average annual sales and all three fees: McDonald's 2025 Franchise Disclosure Document, Items 5, 6 and 19 — 2024 sales across about 12,023 US franchised restaurants open at least a year; median $3.80m, range $914,000 to $19.7m. Rent is shown at 10% for illustration; the FDD discloses effective rents from 0.00% to 33.33% of gross sales.

Three Averages and a Margin

AVERAGE TOTAL COST (ATC)

Total cost ÷ output. What the average meal costs, all in.

AVERAGE VARIABLE COST (AVC)

Variable cost ÷ output. What the average meal costs to actually make.

AVERAGE FIXED COST (AFC)

Fixed cost ÷ output. Overhead spread over the meals. Always falls.

MARGINAL COST (MC)

Change in total cost ÷ change in output. What the NEXT meal costs.

$$ATC = AVC + AFC \text{ and } MC = \Delta TC / \Delta Q$$

Averages tell you how you have done overall. The margin tells you what to do next.

Marginal Cost

Marginal cost is the change in total cost from producing one more unit of output.

MC = ΔTC / ΔQ — what the NEXT meal costs, not what the average one cost

MC FALLS WHILE MP RISES



Early on each worker adds more meals for the same wage, so each meal costs less.

MC RISES WHILE MP FALLS



Later each worker adds fewer meals for the same wage, so each meal costs more.

MC IS A MIRROR OF MP

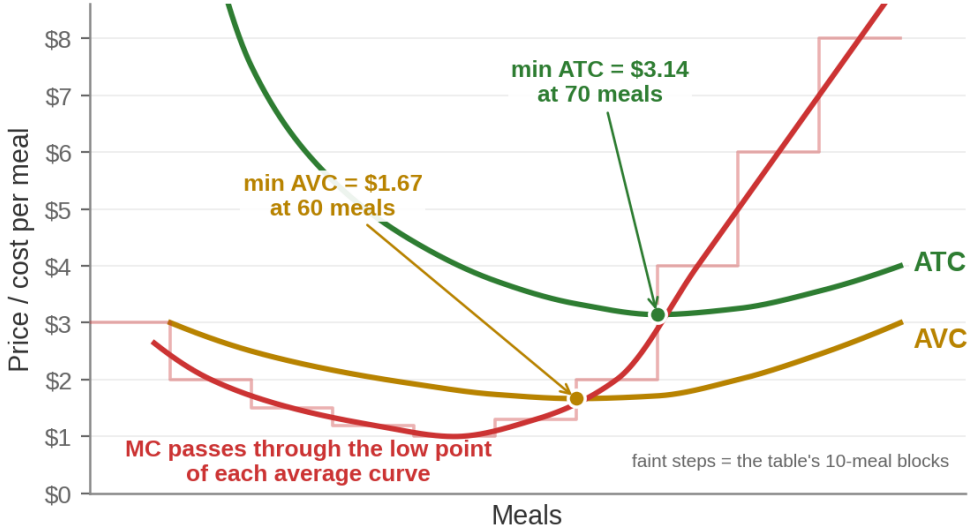


Marginal cost is upside-down marginal product: same story, different units.

The Cost Table

Meals	VC	FC	TC	AVC	AFC	ATC	MC
0	$0	$100	$100	—	—	—	—
10	$30	$100	$130	$3.00	$10.00	$13.00	$3.00
20	$50	$100	$150	$2.50	$5.00	$7.50	$2.00
30	$65	$100	$165	$2.17	$3.33	$5.50	$1.50
40	$77	$100	$177	$1.93	$2.50	$4.42	$1.20
50	$87	$100	$187	$1.74	$2.00	$3.74	$1.00
60	$100	$100	$200	$1.67	$1.67	$3.33	$1.30
70	$120	$100	$220	$1.71	$1.43	$3.14	$2.00
80	$160	$100	$260	$2.00	$1.25	$3.25	$4.00
90	$220	$100	$320	$2.44	$1.11	$3.56	$6.00
100	$300	$100	$400	$3.00	$1.00	$4.00	$8.00

The Cost Curves



Marginal cost passes through the lowest point of each average curve. Always.

The Average Follows the Margin

A margin above the average pulls it up; below, it pulls it down; equal, it is flat.

THE CLASS AVERAGE IS 85%



Somebody joins and scores 100. What happens to the average?

It rises — the new score was above it. Somebody joins and scores 34, and it falls.

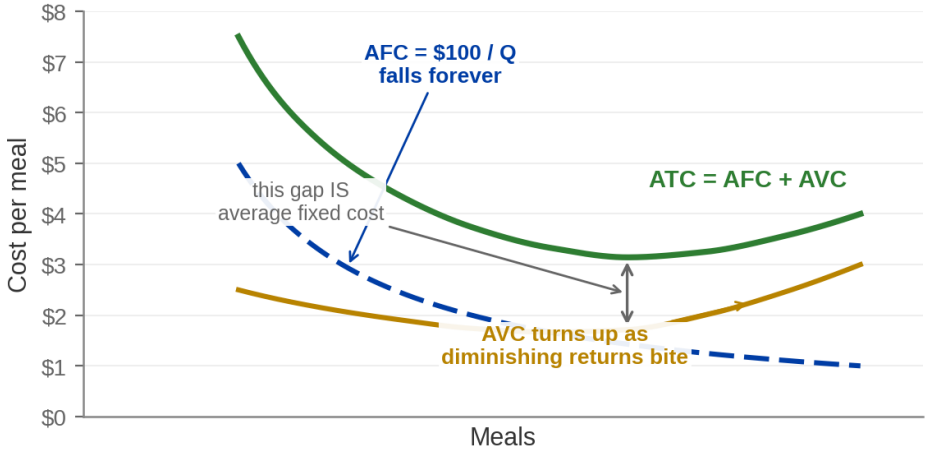
A PLAYER AVERAGES 30 POINTS A GAME



He scores 45 tonight and the average rises. He scores 12 and it falls. He scores exactly 30 and it does not move.

That last case is where MC crosses ATC.

Why the Average Curves Bend



AT A MCDONALD'S

Base rent spreads over more meals, so it falls per meal. The 5% royalty and 4% ad fund are shares of sales — they never spread at all.

AFC FALLS FOREVER

$100 spread over more and more meals. It never turns up — it just flattens toward zero.

ATC IS THE RACE

Falling AFC pulls ATC down; rising AVC pushes it up. AVC wins in the end, and that is the U.

Question 3: How much can you make?

Revenue is the money the firm brings in from selling its product.

Total revenue = price × quantity.

Marginal revenue = ΔTR / ΔQ — what the NEXT meal makes

THE FIRM DOES NOT SET THE PRICE



It is a price taker. The market hands it $4 a meal and that is the end of the negotiation.

SO REVENUE HAS ONE LEVER



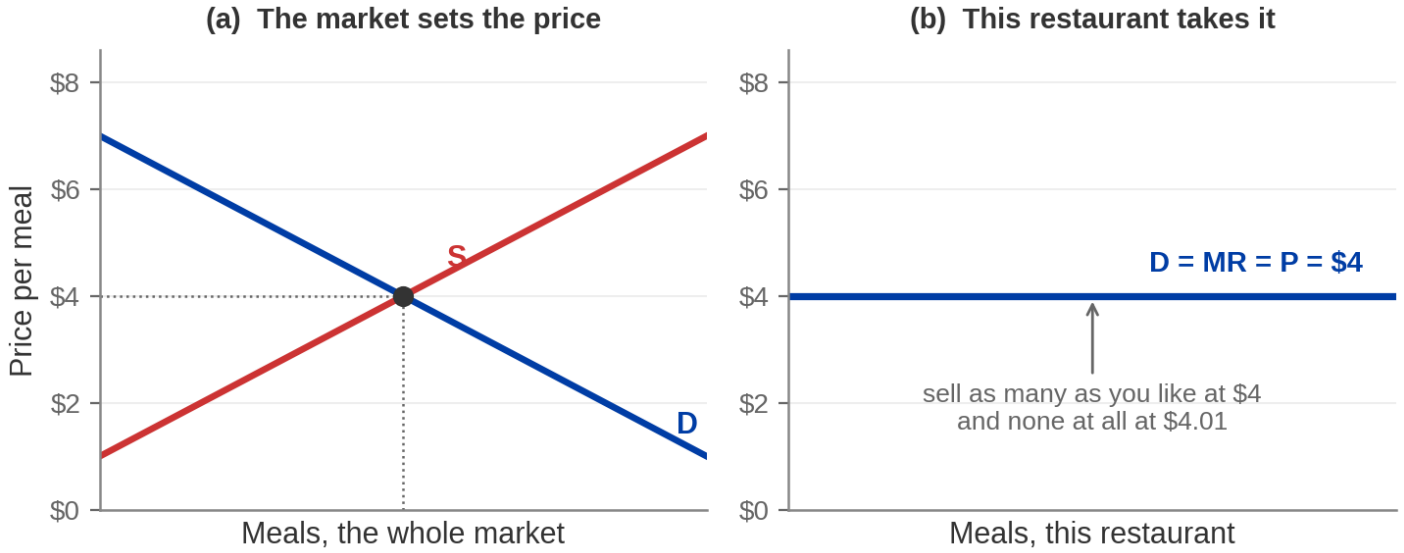
Total revenue = $4 × however many meals it chooses to serve.

AND THE NEXT MEAL ALWAYS ADDS $4



Sell one more, get one more $4. Marginal revenue equals the price.

The Market and the Firm



The Economic Cost of Running a McDonald's

The accountant counts the checks you write. The economist adds what you gave up in order to write them.
Accounting Profit = TR - Explicit costs
Economic Profit = TR – (Explicit cost + Implicit cost)

EXPLICIT — THE INVOICES



Food and paper. Crew. Power. Rent. The 5% royalty. The 4% ad fund. Every one of them is billed, paid, and on the tax return.

IMPLICIT — YOUR TIME



Somebody has to run the restaurant. A US food-service manager earns about $74,900 a year. Do the job yourself and you have given that up.

IMPLICIT — YOUR MONEY



$750,000 of your own cash, locked in for twenty years. At the 4.65% Treasury yield that is $35,000 a year of forgone interest.

Roughly $110,000 a year of real cost that never appears on the tax return.

Food-service manager pay: BLS Occupational Employment and Wage Statistics, May 2025 (mean annual $74,880). Ten-year Treasury constant maturity yield 4.65%, 7 August 2026, US Treasury. Cash requirement: McDonald's franchising costs page.

Walking Toward the Optimum

Keep producing while the next unit brings in more than it costs. Stop when they are equal.

IF MR > MC

The next meal earns more than it costs. Make it. You are leaving money on the table if you stop.

IF MR < MC

The next meal costs more than it earns. Do not make it — and if you already did, make one fewer.

IF MR = MC

Nothing left to gain in either direction. That is the profit-maximising quantity.

The Profit-Maximizing Rule

MR = MC and, because the firm is a price taker, P = MC

STEP 1 — FIND THE QUANTITY

Go along until MR meets MC. At $4 a meal, that is 80 meals an hour.

STEP 2 — READ OFF ATC THERE

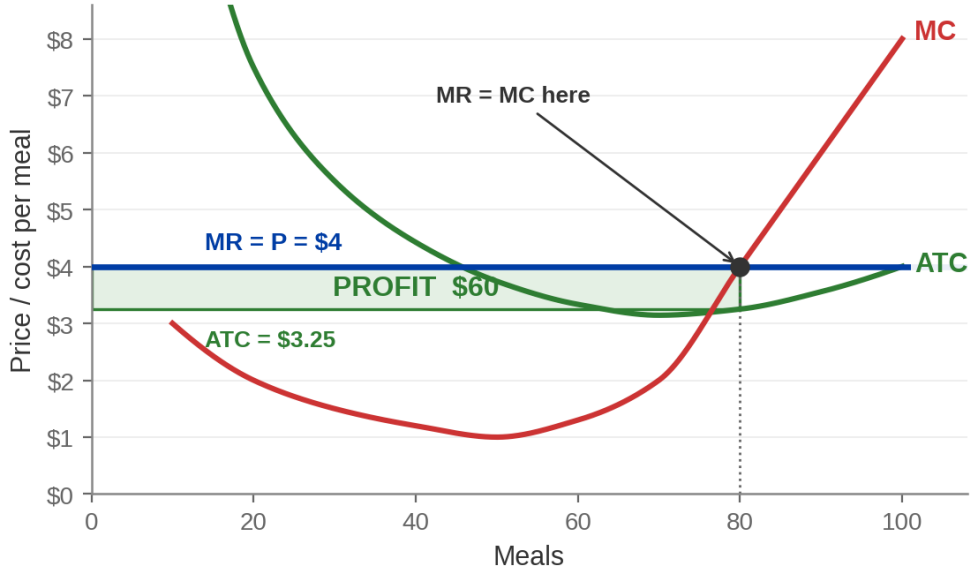
At 80 meals, average total cost is $3.25. That is what each meal cost, all in.

STEP 3 — COMPARE P WITH ATC

P > ATC → profit. P < ATC → loss.
P = ATC → break even, and no reason to be anywhere else.

Three steps. Every profit question in this course is one of them.

Seeing the Profit



PROFIT

Height is P − ATC = $4.00 − $3.25 = $0.75.
Width is 80 meals.
Area is the profit.

TR = 80 × $4 = $320.
TC = $260.
Profit = $60 an hour.

Profit = TR – TC = (P × Q) – (ATC × Q) = (P - ATC) × Q

How a McDonald's Actually Maximizes Profit

MR = MC is not a burger-by-burger decision. In a real restaurant it is an hour-by-hour one.

THE MARGINAL UNIT IS A CREW-HOUR

Roster one more person for one more hour. Does the extra service that hour buys bring in more than the hour costs?

SO WHAT DOES THE HOUR COST?

Fast-food and counter workers average $15.46 an hour nationally. In California the fast-food minimum is $20.00.

AND THE SAME RULE SETS THE HOURS

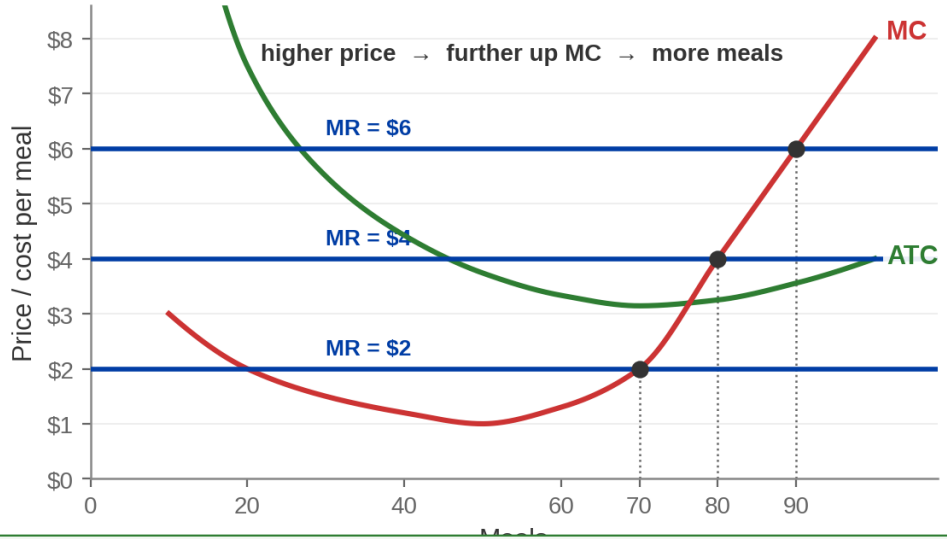
Open at 5am? Staff the second drive-thru lane? Run overnight? Every one of them is MR = MC asked about a smaller unit.

Wages: BLS Occupational Employment and Wage Statistics, May 2025 — fast food and counter workers, mean hourly $15.46, median $15.00. California fast-food minimum wage $20.00 since 1 April 2024 and unchanged in 2026; California general minimum $16.90 from 1 January 2026 (California DIR).

PART 3

From the Seller's Problem to the Supply Curve

Change the Price, Redo the Problem



$2 → 70 meals. $4 → 80 meals. $6 → 90 meals. Three prices, three quantities — and every one of them is a point on MC.

At every price, the firm produces where P = MC. So the MC curve gives quantity supplied at every price unless it shuts down.

Price Elasticity of Supply

The price elasticity of supply measures how responsive quantity supplied is to a change in the market price.

E(S) = % change in quantity supplied ÷ % change in price

ELASTIC: E_S > 1

Quantity is very responsive. A 10% price
rise brings out more than 10% extra.

INELASTIC: E_S < 1

Quantity barely moves. A 10% price rise brings out less than 10% extra.

UNIT-ELASTIC: E_S = 1

They move together. A 10% price rise brings out exactly 10% extra.

Supply elasticity is normally POSITIVE — up with price, unlike demand.

Three Extreme Cases



PERFECTLY ELASTIC

Any amount at $5, nothing at $4.99. Horizontal.

UNIT-ELASTIC

+20% price, +20% quantity. A straight line through the origin.

PERFECTLY INELASTIC

100 units whatever the price. Vertical — a fixed stock, already produced.

What Makes Supply Elastic?

INVENTORY ON HAND

A firm sitting on stock can push it out the moment the price rises. A firm with empty shelves has to make it first.

HOW EASILY IT CAN HIRE

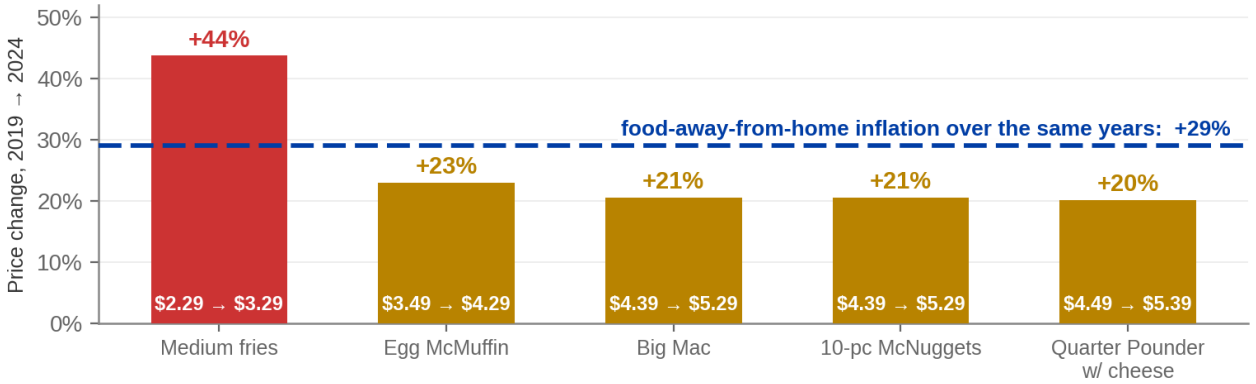
If workers are available at the going wage, output scales quickly. If the skill is scarce, it does not.

TIME — AND THIS IS THE BIG ONE

Over a day, almost nothing can change. Over a year, you can build a second kitchen. Supply is more elastic the longer the horizon.

The same market has a different supply elasticity depending on the window you ask about.

How Elastic Is McDonald's?



SUPPLY ELASTICITY — THIS CHAPTER

Within the hour, roster another crew member: very responsive. Within a year, McDonald's must buy a site and build, and the operator needs 6–12 months of training. The US network grew by 149 restaurants in 2025, on a base of 13,706.

DEMAND ELASTICITY — CHAPTER 5

In the second quarter of 2026 US comparable sales rose 0.8% — on a bigger average check, “partly offset by negative comparable guest counts”. Prices up, visits down.

Menu prices: McDonald's USA, “McD US Pricing Myths vs Facts”, 29 May 2024, with its own 29% food-away-from-home benchmark. Comparable sales and guest counts: McDonald's Q2 2026 earnings release, 4 August 2026. Restaurant counts: McDonald's FY2025 Form 10-K. Published elasticity for food away from home: −0.81 (Andreyeva, Long & Brownell, AJPH 100(2), 2010).

Stay Open or Shut Down: What Do You Still Pay?

STAY OPEN

You pay your VARIABLE costs — the crew, the beef, the power — and you pay your FIXED costs.

But you also take in revenue.

SHUT DOWN

You pay no variable costs. Nobody is rostered, nothing is bought.

You still pay every penny of your FIXED costs. And you take in nothing.

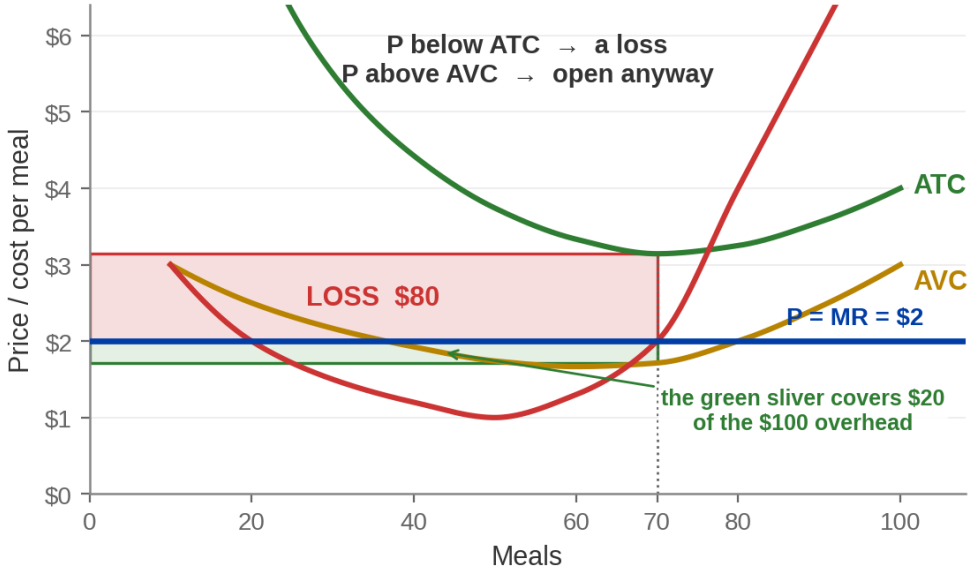


Fixed costs are paid in BOTH columns.

AT A MCDONALD'S

Open at 4am? Run overnight? The rent is owed either way; the crew, the power and the food are not.

Losing Money and Still Opening the Doors



SHUT DOWN

Revenue $0.
Variable cost $0.
Fixed cost $100.
Lose the whole $100.

STAY OPEN AT $2

Revenue $140.
Variable cost $120.
Fixed cost $100.
Lose $80 — which is $20 better.

Sunk Costs

A sunk cost has already been committed and can never be recovered. It should not affect any decision you make from now on.

IN THE SHORT RUN, FIXED COSTS ARE SUNK



The lease is signed, the fryers are bought. That money is gone whatever you decide tonight.

SO THEY DROP OUT OF THE DECISION



Not because they do not matter — they matter enormously — but because they are the same under every option.

THIS IS NOT JUST A FIRM PROBLEM

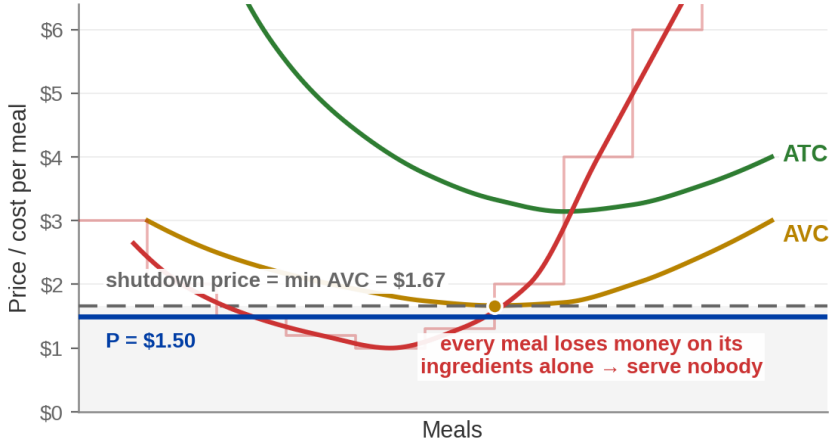


The ticket you already bought. The two years already spent on the degree. The meal already paid for. Same logic, same error.

"But I've already spent so much on it" is never, by itself, a reason to continue.
If we have a reason to stop producing, sunk cost should not be stopping us from shutting down.

The Shutdown Rule

Shut down if P < AVC. Keep producing if P > AVC.



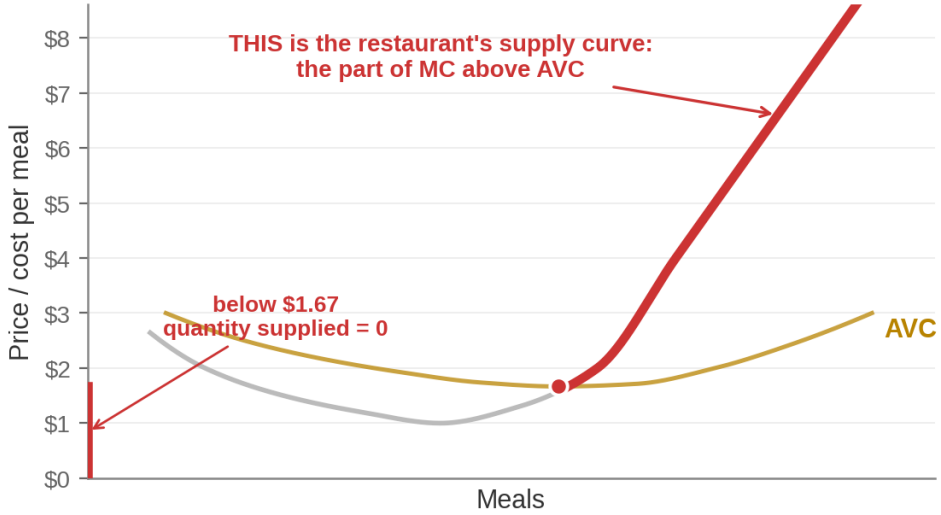
BELOW $1.67

Every meal loses money on its ingredients alone. Serving makes things worse. Serve nobody.

ABOVE $1.67

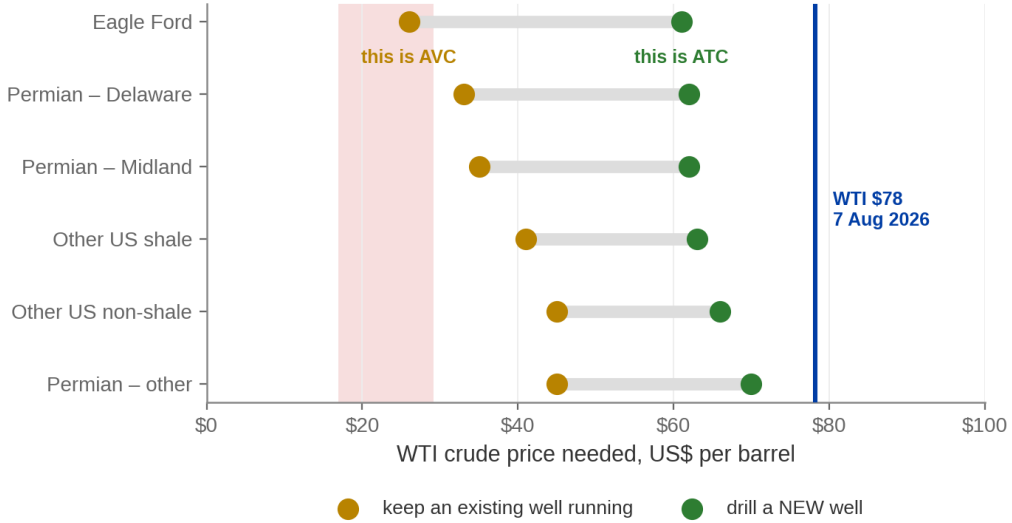
Every meal contributes something toward the overhead. Open, even at a loss.

The Firm's Supply Curve



The short-run supply curve is the portion of the marginal cost curve that lies above average variable cost.

The Shutdown Rule in the Oil Patch





Existing wells keep running down to $26–$45. A NEW well needs $61–$70. The gap between them is AVC against ATC.

Dallas Fed Energy Survey special questions, fielded 12–20 March 2025, published 26 March 2025 (n = 83 and 81). The March 2026 round moved the all-respondent averages to $43 and $66. WTI spot $78.18 on 7 August 2026.

PART 4

Producer Surplus

Producer Surplus

Producer surplus is the difference between the price a seller actually receives and the lowest price it would have accepted.

THE MIRROR OF CHAPTER 5

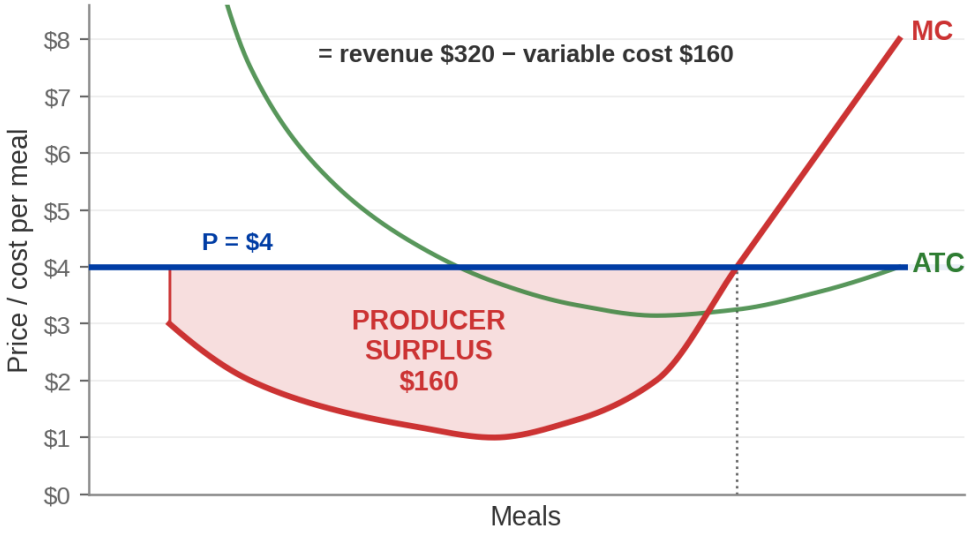
Consumer surplus was the gap between what a buyer WOULD have paid and what she did pay — the area under demand, above price.

PRODUCER SURPLUS

The gap between what a seller receives and what it would have accepted — the area above supply, below price.

The supply curve is a marginal cost curve, so "lowest acceptable price" means marginal cost.

Measuring Producer Surplus



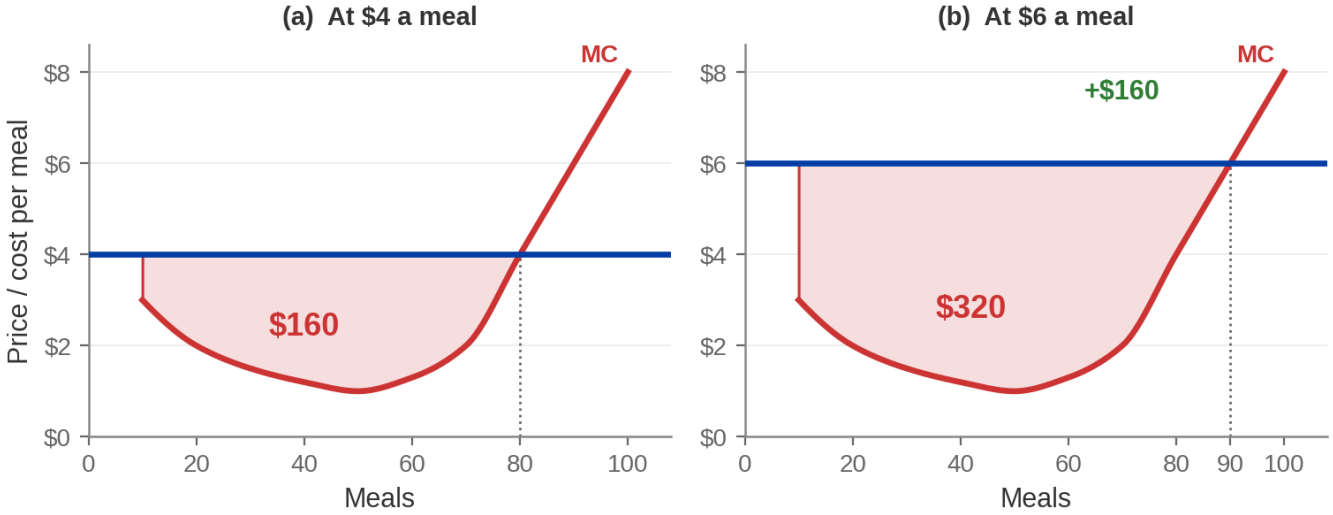
MEAL BY MEAL

The 10th meal would have been served for $3 and earns $4. The 80th costs exactly $4 and earns $4 — no surplus at all.

NOT THE SAME AS PROFIT

Producer surplus $160.
Profit $60.
The gap is exactly the $100 of fixed cost.

What a Price Rise Adds



$4 → $6 takes producer surplus from $160 to $320. Sellers gain $160 — and buyers lose.

PART 5

From the Short Run to the Long Run

In the Long Run, Everything Is Variable

SHORT RUN

Some factors fixed. The only way to change output is to change labor/utilities/food— more people in the same kitchen.

That is why diminishing returns bites, and why MC eventually rises steeply.

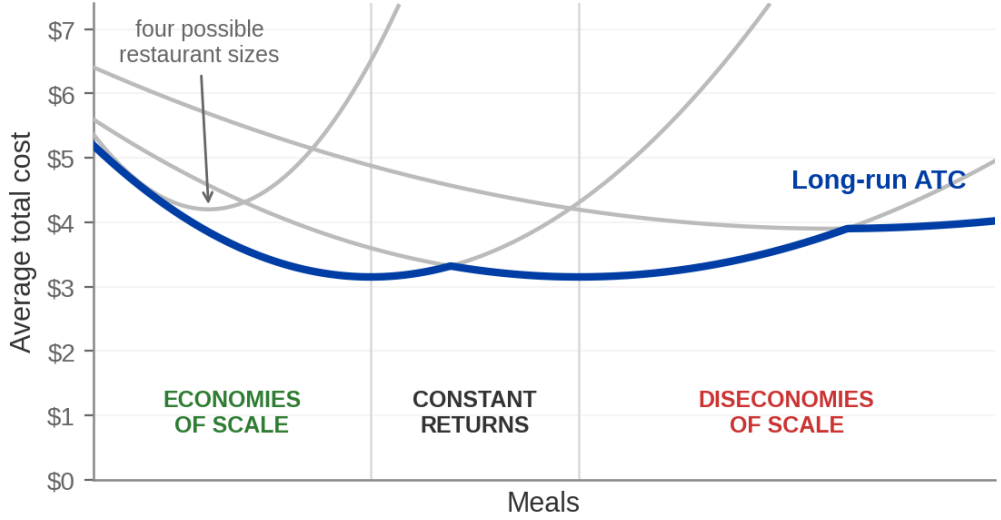
LONG RUN

Nothing fixed. To change output you can change ANYTHING — take the unit next door, add a second line, move.

The long run is a planning period: what is the right SIZE of restaurant?

In the short run you ask "how much should I make?" In the long run you ask "how big should I be?"

The Long-Run Average Cost Curve



Each grey curve is one possible restaurant size. The blue curve is the cheapest you can do at each output, once you can choose the size.

Economies, Constant Returns, Diseconomies

ECONOMIES OF SCALE

ATC FALLS as output rises. Double the inputs, more than double the output.

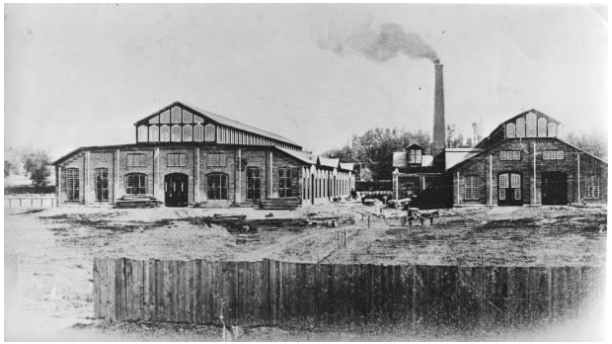
Why: large set-up costs get spread, and bigger operations specialise further.



CONSTANT RETURNS TO SCALE

ATC is FLAT. Double the inputs, double the output.

Why: the gains from specialising have all been taken, and nothing has started going wrong yet.



DISECONOMIES OF SCALE

ATC RISES. Double the inputs, less than double the output.

Why: coordination. Too many layers of management, too far between the decision and the work.



Profits Are a Signal — and Everybody Can See Them

In the long run, firms can enter an industry or leave it. Nothing legal or technical stands in the way.

THE FRANCHISE BEATS THE TREASURY

Put the $2.1 million a restaurant costs into 10-year Treasuries at 4.65% and it returns about $98,000 a year, risk-free. Clear more than that and the difference is economic profit.

SO MORE MCDONALD'S OPEN

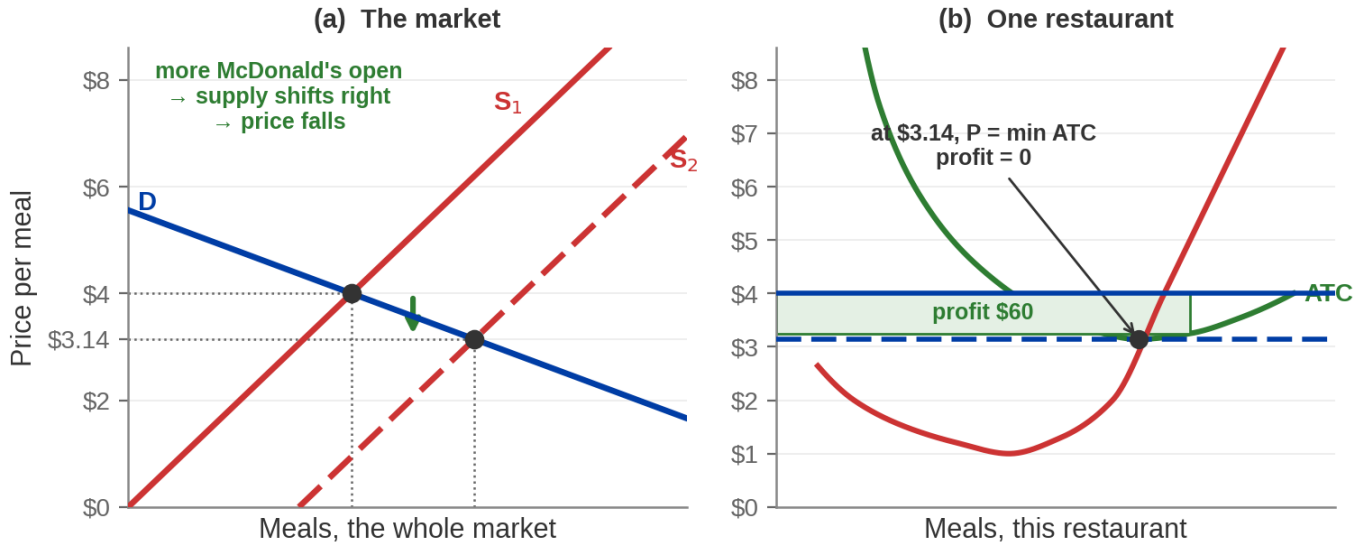
Anybody who can read the disclosure document can see the same numbers. Restaurants get built, market supply shifts right — and a supply curve shifting right means the price comes down.

WHICH SHRINKS THE VERY PROFIT THAT ATTRACTED THEM

Entry continues while there is still an advantage over the Treasury. It stops when there is not.

The profit is not destroyed by bad luck. It is destroyed by the response to it.

Entry Competes the Profit Away



Entry stops when P = minimum ATC = $3.14, because at that price there is nothing left to attract anybody.

Zero Economic Profit Is Not Zero Profit

In long-run equilibrium, price equals minimum average total cost and economic profit is zero.

WHAT IT DOES NOT MEAN



It does not mean the owner is broke, working for free, or should quit. She is being paid, and her capital is earning its normal return.

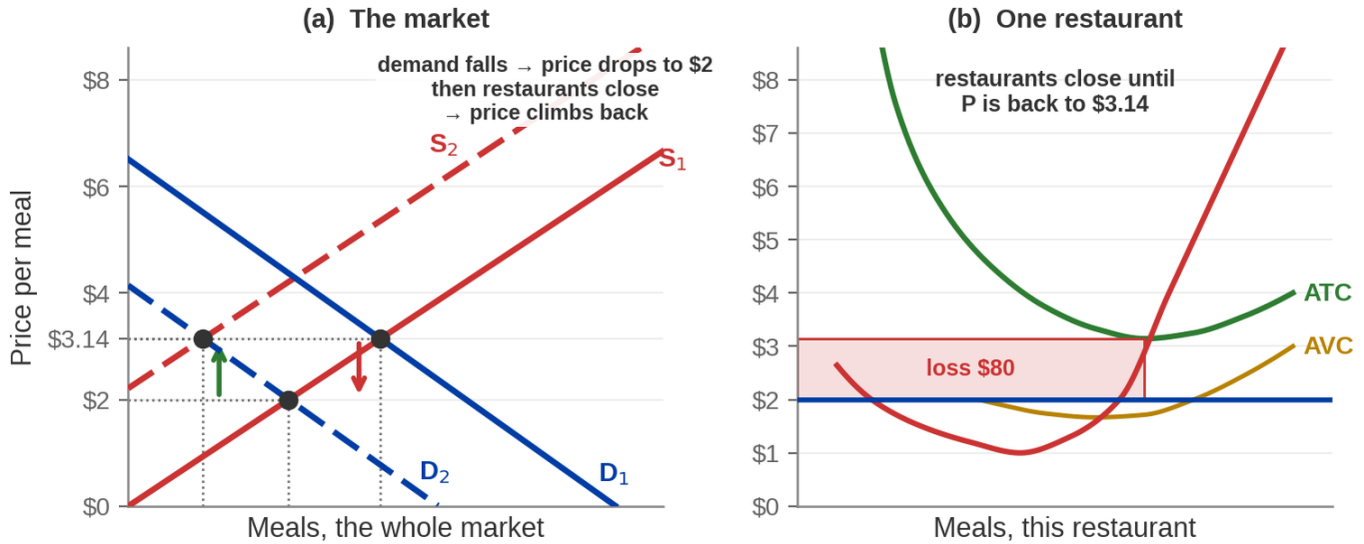
WHAT IT DOES MEAN



There is nothing better available. Every alternative use of the same time, money and space returns exactly the same. She has no reason to move — and neither does anyone outside.

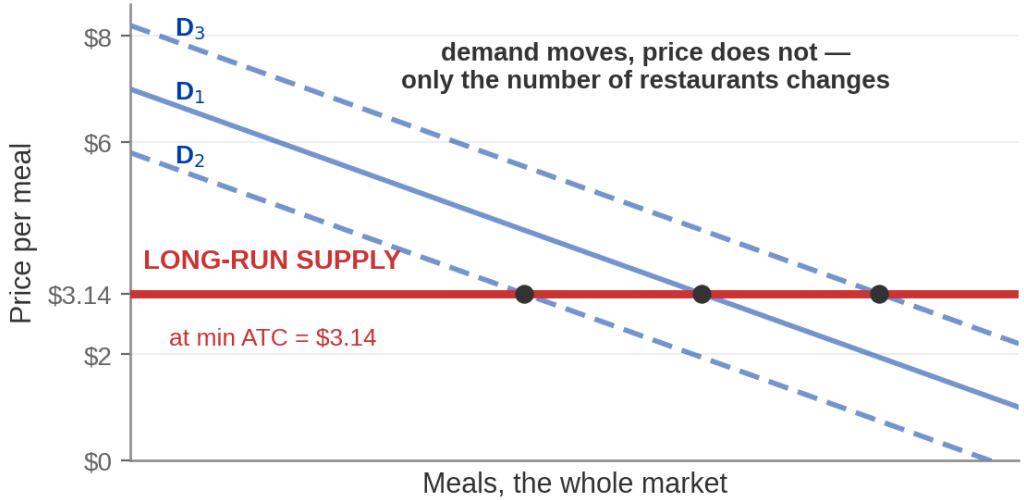
"Zero economic profit" means "doing exactly as well as the next best thing" — which is fine.

And Exit Works the Same Way



Demand falls, price drops to $2, restaurants lose money, some close — and the price climbs back to $3.14.

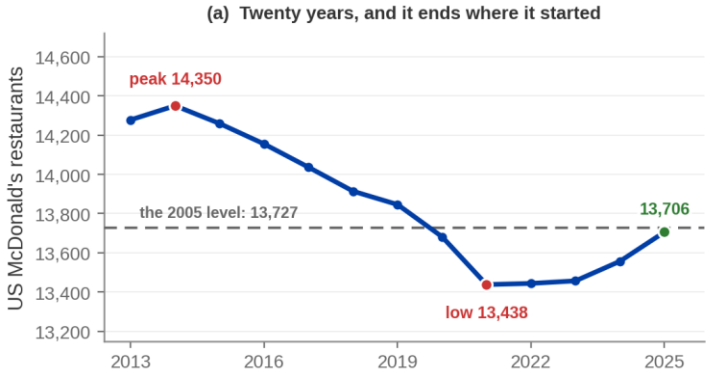
Why Long-Run Supply Is Flat

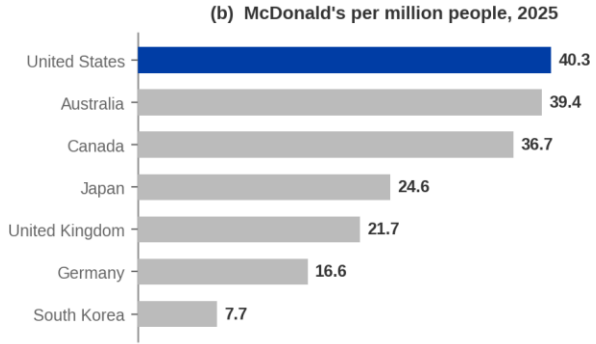


In the long run the price returns to minimum ATC whatever demand does. What changes is the NUMBER of firms.

And $3.14 is the BOTTOM of the ATC curve — so entry pushes every restaurant to its efficient scale. Nobody planned that.

So How Many McDonald's Does America End Up With?





TWENTY YEARS, NET MINUS TWENTY-ONE

13,727 US restaurants in 2005. 13,706 in 2025. It peaked at 14,350 in 2014 and bottomed at 13,438 in 2021 — seven straight years of net closures.

ENTRY WENT SOMEWHERE ELSE

Of ~2,600 restaurants planned worldwide for 2026, about 750 are in the US and all International Operated Markets. Over 1,800 are in licensed markets.

Nobody chose 13,706. Entry chose it — and then it stopped.

US counts: McDonald's Corp — Q4 2005 earnings release, Q4 releases 2015–2018, and Forms 10-K FY2020/FY2022/FY2024/FY2025. 2026 plan: FY2025 10-K, MD&A. Country counts: “Restaurants by Market 2025”. Populations: national statistical offices; US Census Bureau Vintage 2025 (340,003,797 at 1 Jul 2025).

Chapter 6: The Full Picture

	What it says	Key terms	Where you saw it
Production	Output rises with labour, then more slowly, then falls. The kitchen is fixed in the short run.	marginal product, diminishing returns, specialisation	The 3rd worker adds 15
Costs	TC = VC + FC. The average follows the margin, so MC passes through the low point of each average curve.	FC, VC, ATC, AVC, AFC, MC	min AVC $1.67, min ATC $3.14
The rule	Produce where MR = MC, and MR = P for a price taker. Then compare P with ATC.	price taker, economic profit, implicit cost	$4 → 80 meals → $60
Supply	Short-run supply is MC above AVC. Below minimum AVC, produce nothing.	shutdown rule, sunk cost, ε_S	$2 → lose $80, not $100
Surplus	The area above supply and below price — it exceeds profit by the fixed cost.	producer surplus, willingness to accept	$160 surplus, $60 profit
The long run	Entry and exit drive price to minimum ATC, so long-run supply is flat and profit is zero.	economies of scale, free entry, long-run equilibrium	Nick, Homer, and ethanol

Chapter 5 built the demand curve. Chapter 6 built the supply curve. Chapter 4's diagram is now derived from both sides.

KEY TAKEAWAYS

Chapter 6: Sellers and Incentives

- The average follows the margin — which is why MC passes through the bottom of ATC.
- Produce where MR = MC. For a price taker MR = P, so produce where P = MC.
- Compare P with ATC for profit; compare P with AVC for whether to open at all.
- Fixed costs are sunk in the short run. They cannot decide anything.
- Free entry drives price to minimum ATC — and economic profit to zero.